

Executive Summary: An optimized hydraulic sizing and condensation analysis was performed for the **Natural Gas Boiler Supply Line (Optimization - Case 1 - R2)** at an optimized total throughput of **571.8 kg/h** (32.08 kgmol/h). The line diameter for the main run (PIPE-01 and PIPE-02) is optimized to **DN 80 (3" Sch 40, ID: 77.92 mm)** downstream of an SSV station. The network was evaluated under **Cold Ambient Condition** (condensate formation check) and **Hot Ambient Condition** (high temperature check). Both scenarios verify 100% vapor fraction (zero liquid dropout), low line pressure drops ($\Delta P \leq 0.156$ bar total), and dependable delivery pressures of ~3.84 to 3.85 bar(g) at both boiler burners.

1. DESIGN BASIS & OPERATING CONDITIONS

Natural gas is received at 4.000 bar(g) and 46.00°C from battery limits (NG FEED) at an optimized flow rate of 571.8 kg/h. The gas passes through an automated Slam Shut Valve station (SSV STATION) with an engineered safety pressure loss allowance of 0.050 bar. The main distribution piping rises +5.5 m elevation along PIPE-02 before entering the distribution manifold (BRANCH HEADER), splitting equally into two 285.9 kg/h streams feeding Boiler 1 and Boiler 2.

PARAMETER	UNIT	NG FEED SOURCE	SSV STATION	BOILER 1 DEMAND	BOILER 2 DEMAND
Mass Flow Rate	kg/h	571.8	571.8	285.9	285.9
Operating Temperature	°C	46.00	Isenthalpic Flow	Floating (Heat Exch.)	Floating (Heat Exch.)
Operating Pressure	bar(g)	4.000	3.949	~3.844 – 3.851	~3.844 – 3.850
Molar Flow	kgmol/h	32.08	32.08	16.04	16.04
Fluid Phase	-	Vapor (Frac: 1.0)	Vapor (Frac: 1.0)	Vapor (Frac: 1.0)	Vapor (Frac: 1.0)

2. COMPARATIVE SIMULATION RESULTS (NODE BY NODE)

NODE / STREAM DESCRIPTION	COLD AMBIENT CASE (CONDENSATE CHECK)			HOT AMBIENT CASE (HIGH TEMP CHECK)		
	PRESS [BAR(G)]	TEMP [°C]	VAP FRAC / DROPOUT	PRESS [BAR(G)]	TEMP [°C]	VAP FRAC / DROPOUT
NG FEED (Inlet Battery Limit)	4.000	46.00	1.0000 (0 kg/h)	4.000	46.00	1.0000 (0 kg/h)
OUT 1 (SSV Station Inlet)	3.999	45.85	1.0000 (0 kg/h)	3.999	46.00	1.0000 (0 kg/h)
SSV OUT (SSV Station Outlet)	3.949	45.82	1.0000 (0 kg/h)	3.949	45.98	1.0000 (0 kg/h)
HEADER OUT (Manifold Entry)	3.855	18.37	1.0000 (0 kg/h)	3.849	45.98	1.0000 (0 kg/h)
BOILER 1 - IN (Target Consumer 1)	3.851	17.93	1.0000 (0 kg/h)	3.844	45.98	1.0000 (0 kg/h)
BOILER 2 - IN (Target Consumer 2)	3.850	17.42	1.0000 (0 kg/h)	3.844	45.98	1.0000 (0 kg/h)

3. PIPING SEGMENT HYDRAULIC SUMMARY (OPTIMIZED DIAMETERS)

PIPE SEGMENT ID	OPTIMIZED DIMENSIONS (ID × LENGTH)	ELEVATION	ΔP COLD (BAR)	ΔP HOT (BAR)	HEAT DUTY	CONDENSATE DROPOUT
PIPE-01	DN 80 (77.92 mm ID, 3") × 0.50 m	0.0 m	0.001	0.001	E1 (+191.6 / -0.19 kJ/h)	0.0000 kg/h
PIPE-02	DN 80 (77.92 mm ID, 3") × 7.50 m	+5.5 m	0.094	0.100	E2 (+33,580 / -70.4 kJ/h)	0.0000 kg/h
PIPE-201 (Boiler 1 Leg)	DN 50 (52.50 mm ID, 2") × 4.50 m	0.0 m	0.004	0.005	B-E1 (+264.7 / -0.57 kJ/h)	0.0000 kg/h
PIPE-202 (Boiler 2 Leg)	DN 80 (77.93 mm ID, 3") × 5.00 m	0.0 m	0.005	0.005	B-E2 (+573.2 / -2.40 kJ/h)	0.0000 kg/h

4. ENGINEERING OBSERVATIONS & TECHNICAL RECOMMENDATIONS

HYDRAULIC PERFORMANCE • Sizing Optimization: Main line size reduced from DN 100 to DN 80 (3"), optimizing piping material costs and installation footprint while maintaining total frictional losses at ≤ 0.106 bar. • Burner Arrival Pressure: Delivery pressure remains strong at ~3.84 to 3.85 bar(g), providing ample control cushion for boiler burner trains. • Flow Sizing Margin: The 571.8 kg/h flow is supported within optimal velocity limits (avoiding noise and vibration).	THERMAL & CONDENSATION INTEGRITY • Inlet Condition: Feed arrives at 46.00°C and 4.00 bar(g). • Ambient Influence: ◦ Cold Case: Convective heat loss along uninsulated piping drops gas to 17.42°C – 17.93°C at boiler limits. ◦ Hot Case: Gas temperature remains virtually isothermal at 45.98°C. • Dew Point Margin: Vapor fraction is strictly 1.0000; no condensation dropout occurs in either case.
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5. DESIGN COMPLIANCE & CONCLUSION

The **Optimization - Case 1 (R2)** design is verified as hydraulically and thermodynamically viable. The line reduction to **DN 80 (3")** for the main distribution segment achieves substantial capital cost reduction while comfortably maintaining delivery pressure at **~3.84 – 3.85 bar(g)** across both cold winter and hot summer conditions. Single-phase vapor flow is guaranteed throughout.